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Revealing invisible flicker by Fourier Transform

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50×××××

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学文



Instructor Jun Chen

Date May.2025

Introduction



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DJI ACTION 4



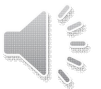
Resolution:**1080P**

Frame rate:**240**

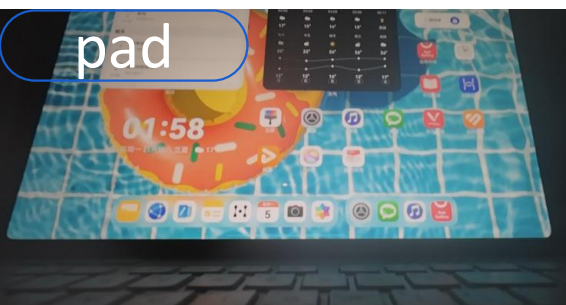
MATLAB



vesion:**R2024b**



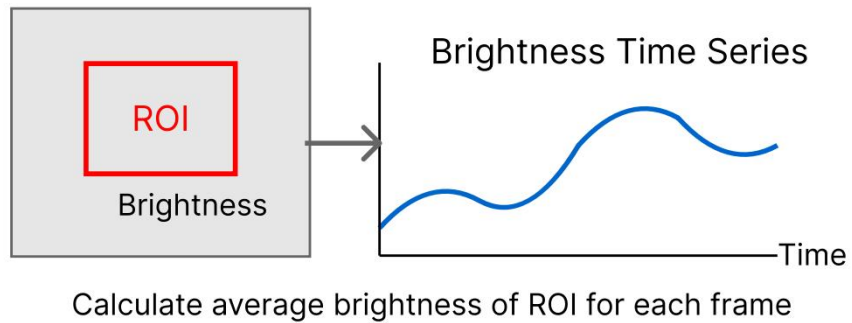
Record videos



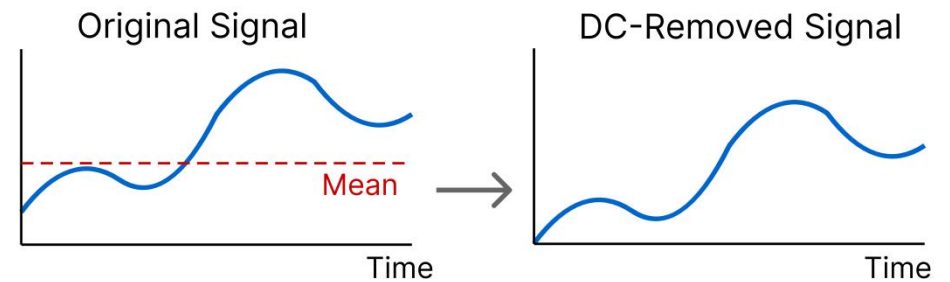
 run.m 类型: Objective C 源文件	修改日期: 2025/5/5 23:13 大小: 2.78 KB
 pad120hz_on.MP4 时长: 00:00:06	帧高度: 1080 帧宽度: 1920 修改日期: 2025/5/5 4:18 大小: 66.4 MB
 pad120hz_on(2).MP4 时长: 00:00:05	帧高度: 1080 帧宽度: 1920 修改日期: 2025/5/5 1:44 大小: 73.6 MB
 pad120hz_on (2).MP4 时长: 00:00:06	帧高度: 1080 帧宽度: 1920 修改日期: 2025/5/5 3:28 大小: 56.2 MB
 pad120hz_off.MP4 时长: 00:00:06	帧高度: 1080 帧宽度: 1920 修改日期: 2025/5/5 1:58 大小: 54.5 MB
 pad60hz_on.MP4 时长: 00:00:05	帧高度: 1080 帧宽度: 1920 修改日期: 2025/5/5 1:09 大小: 76.8 MB
 pad60hz_off.mp4 时长: 00:00:07	帧高度: 1080 帧宽度: 1920 修改日期: 2025/5/4 22:56 大小: 93.5 MB
 moving90hz.mp4 时长: 00:00:07	帧高度: 1080 帧宽度: 1920 修改日期: 2025/5/5 18:47 大小: 66.4 MB
 Light_flicker_detection_using_high-speed_imaging.pdf	修改日期: 2025/5/4 20:29 大小: 768 KB
 light_f_detection.m 类型: Objective C 源文件	修改日期: 2025/5/5 22:53 大小: 16.8 KB
 light.mp4 时长: 00:00:06	帧高度: 1080 帧宽度: 1920 修改日期: 2025/5/4 20:52 大小: 90.2 MB
 library_laptop60hz_on.mp4 时长: 00:00:06	帧高度: 1080 帧宽度: 1920 修改日期: 2025/5/5 17:46 大小: 59.0 MB
 laptop144hz_on.MP4 时长: 00:00:06	帧高度: 1080 帧宽度: 1920 修改日期: 2025/5/4 23:30 大小: 92.0 MB
 laptop144hz_off.mp4	帧高度: 1080 修改日期: 2025/5/4 21:45

Mathematical Principles

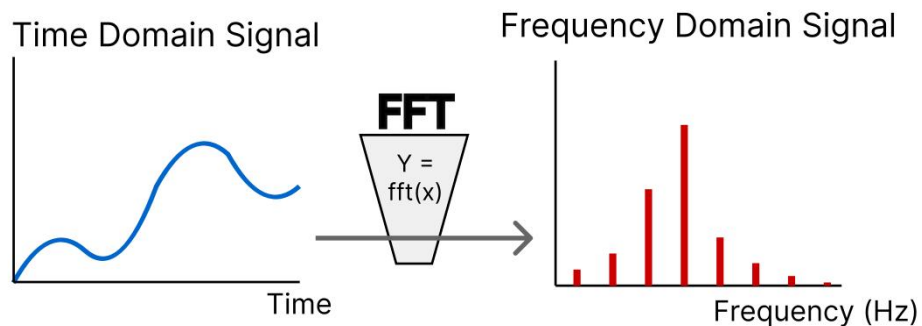
Step 1: Signal Acquisition



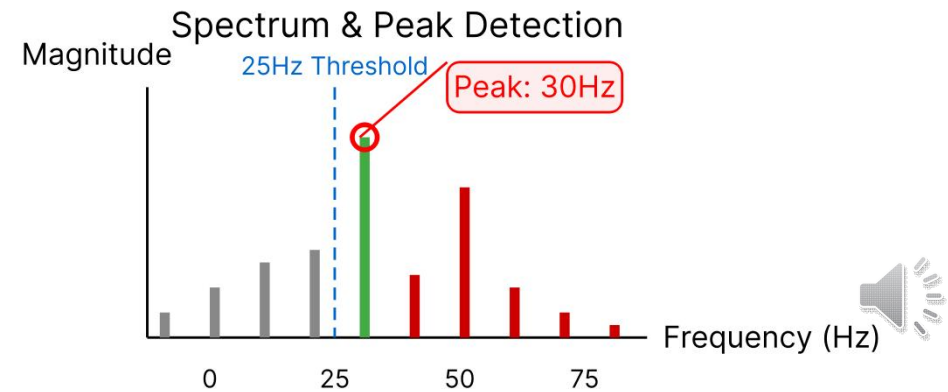
Step 2: DC Component Removal



Step 3: Fast Fourier Transform (FFT)



Step 4: Peak Frequency Detection



MatlabCodes

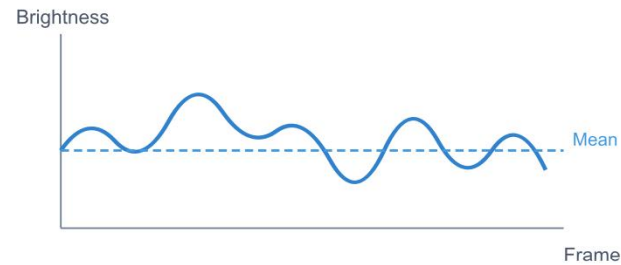
```

1 % Convert the script to a function, remove default parameters at the top, and use input variable
2 function light_f_detection(video_file, output_folder, N_frames, frame_rate, analysis_dimension,
3
4 % Modify the output folder naming rule to frames_videoName_frameCount
5 [~, video_name, ~] = fileparts(video_file);
6 % Create a display name for the video (replace underscores with hyphens to avoid subscript form
7 display_video_name = strrep(video_name, '_', '-');
8 output_folder = fullfile(fileparts(output_folder), sprintf('frames %s %d', video_name, N_frames));
9
10 % Create a corresponding results folder
11 % Extract folder number from output_folder
12 [~, folder_name, ~] = fileparts(output_folder);
13 folder_number = '';
14 for i = 1:length(folder_name)
15     if isstrprop(folder_name(i), 'digit')
16         folder_number = [folder_number, folder_name(i)];
17     end
18 end
19
20 % Create a results folder with the corresponding number
21 results_folder = fullfile('results', [ 'results', folder_number]);
22 if ~exist(results_folder, 'dir')
23     mkdir(results_folder);
24 end
25 fprintf('Results will be saved to folder: %s\n', results_folder);
26
27 % --- Step 1 & 2: Video frame extraction ---
28 % Check if the video file exists
29 if ~exist(video_file, 'file')
30     error('Video file "%s" does not exist. Please ensure it is in the same directory as the scr
31 end
32
33 if ~exist(output_folder, 'dir')
34     mkdir(output_folder);
35 end

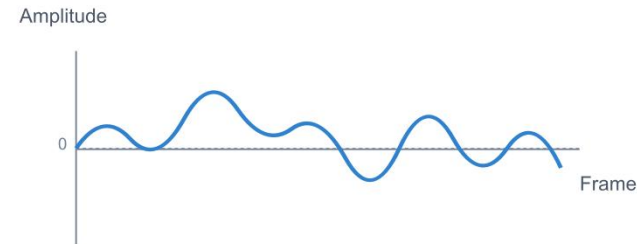
```

1D Flicker Analysis: Signal Processing & MATLAB Implementation

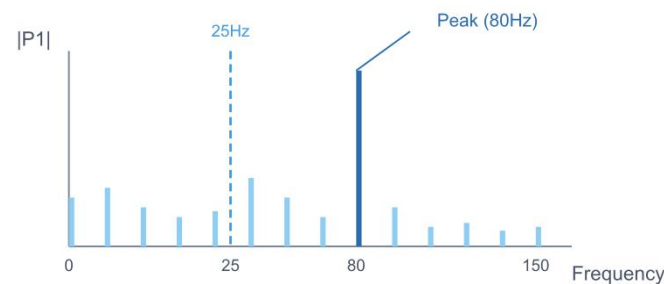
Original Time Signal



DC Removed Signal



Amplitude Spectrum



MATLAB Implementation

```

% 1. Extract Frames & ROI Avg. Brightness
for i = 1:N_frames
    roi = imcrop(frames{i}, roi_rect);
    brightness_signal_1d(i) = mean(roi(:));
end

% 2. Remove DC Component
brightness_signal_1d = brightness_signal_1d -
    mean(brightness_signal_1d);

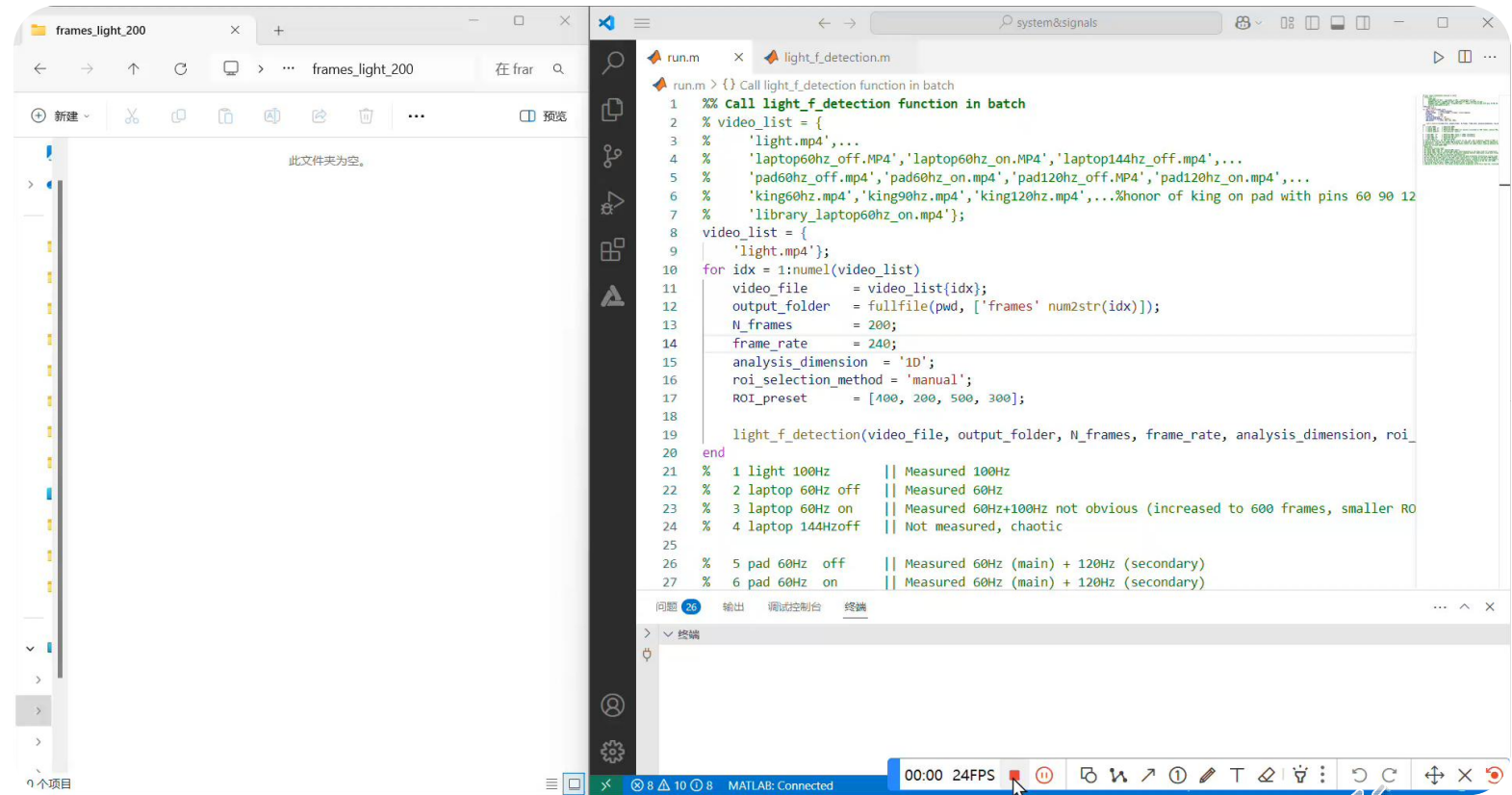
% 3. FFT & Calculate Magnitude Spectrum
Y = fft(brightness_signal_1d);
L = N_frames;
P2 = abs(Y/L);
P1 = P2(1:floor(L/2)+1);
P1(2:end-1) = 2*P1(2:end-1);
f_axis_1d = frame_rate*(0:(L/2))/L;

% 4. Find Peak Frequency (≥25Hz)
freq_mask = f_axis_1d >= 25;
[max_amp, idx] = max(P1(freq_mask));

```



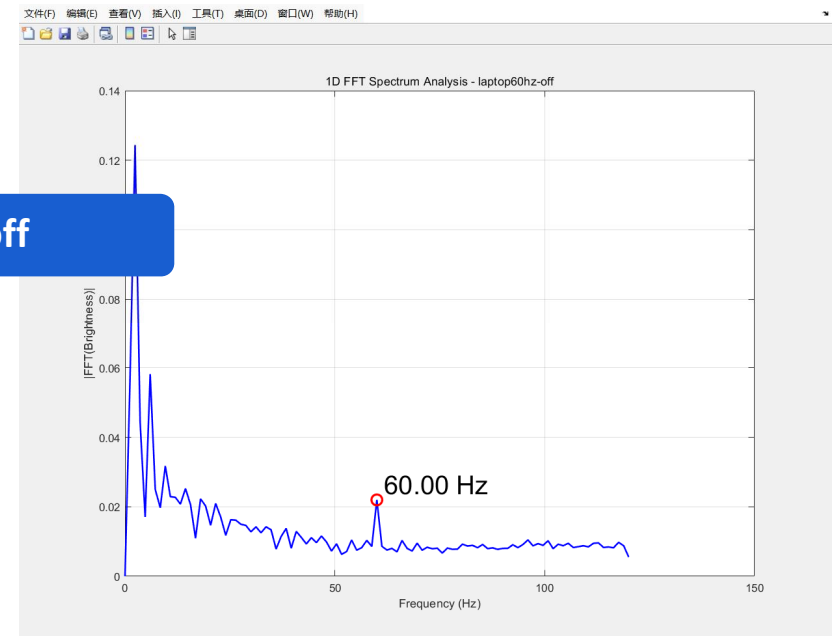
Process the video



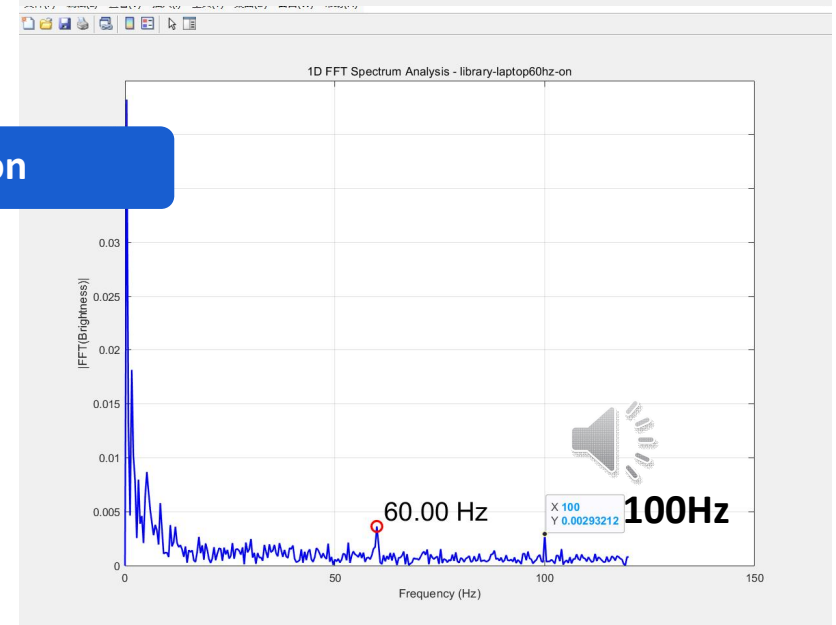
Baseline Validation

Results Analysis

laptop_60hz_off



laptop_60hz_on



Results Analysis



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Nyquist–Shannon sampling theorem

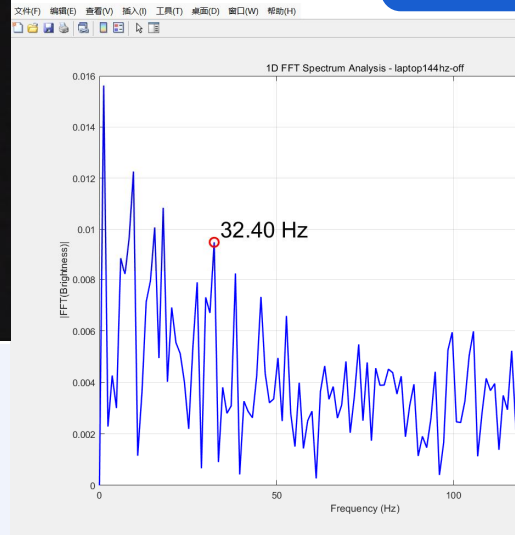
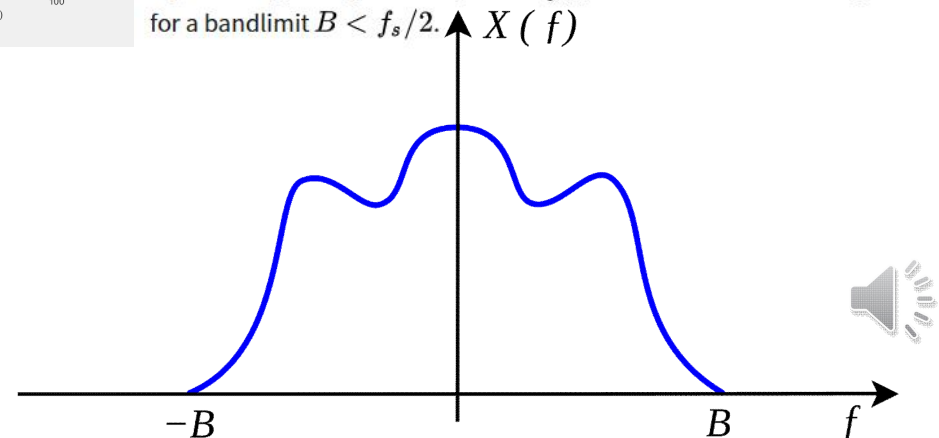
Introduction [edit]

Sampling is a process of converting a signal (for example, a function of continuous time or space) into a sequence of values (a function of discrete time or space).

Shannon's version of the theorem states:[2]

Theorem—If a function $x(t)$ contains no frequencies higher than B hertz, then it can be completely determined from its ordinates at a sequence of points spaced less than $1/(2B)$ seconds apart.

A sufficient sample-rate is therefore anything larger than $2B$ samples per second. Equivalently, for a given sample rate f_s , perfect reconstruction is guaranteed possible for a bandlimit $B < f_s/2$.



显示器信息

B156HAN15.H
显示器 1: 已连接到 Intel(R) Iris(R) Xe Graphics

桌面模式 1920 × 1080, 144.06 Hz
活动信号模式 1920 × 1080, 144.06 Hz
位深度 8 位
颜色格式 RGB
颜色空间 标准动态范围(SDR)

显示器 1 的显示适配器属性

选择刷新率
较高的速率可提供更流畅的动作,但也消耗更多的电源电量 有关刷新率的详细信息

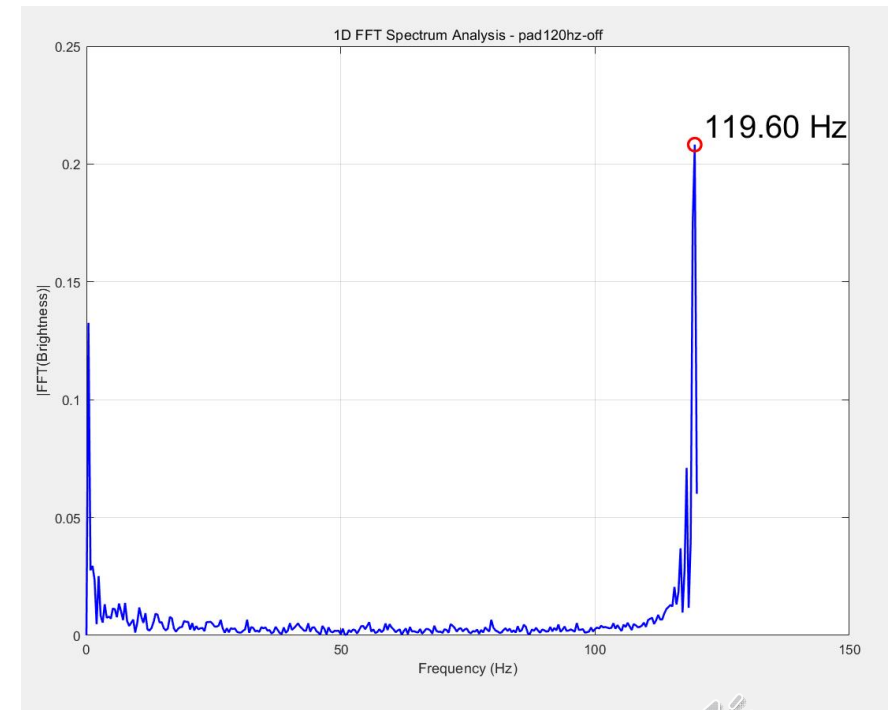
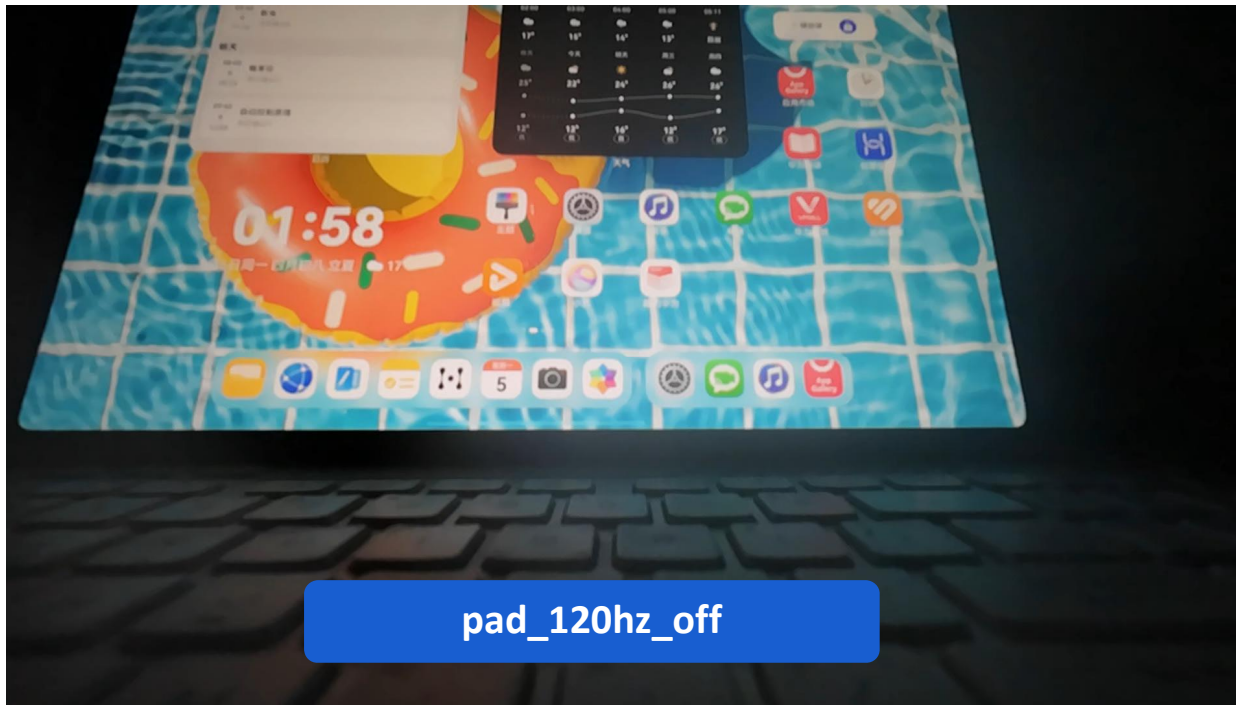
144.06 Hz
60.06 Hz

自动管理应用的颜色
自动颜色管理可确保你的应用和其他内容在此显示器上具有准确的颜色(这可能会消耗更多电力) 有关自动颜色管理的详细信息

laptop 144hz

Results Analysis

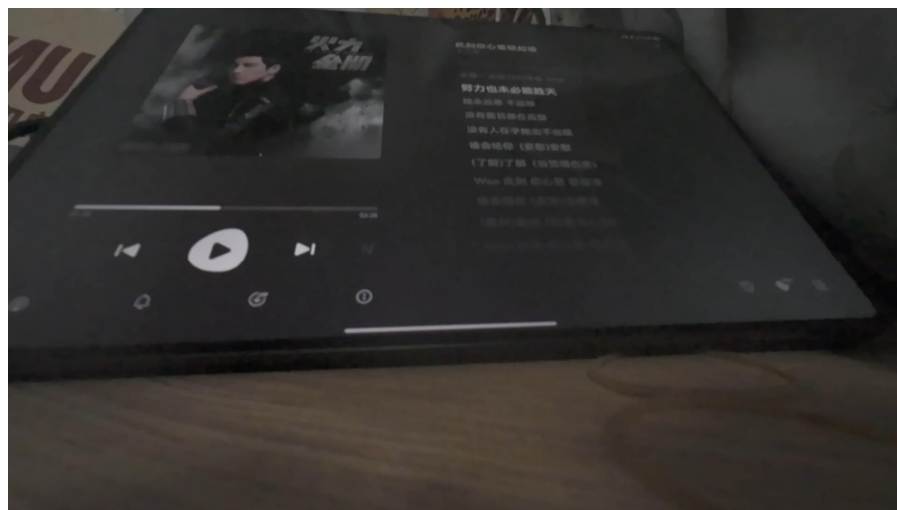
Frequency of Extreme Measurement



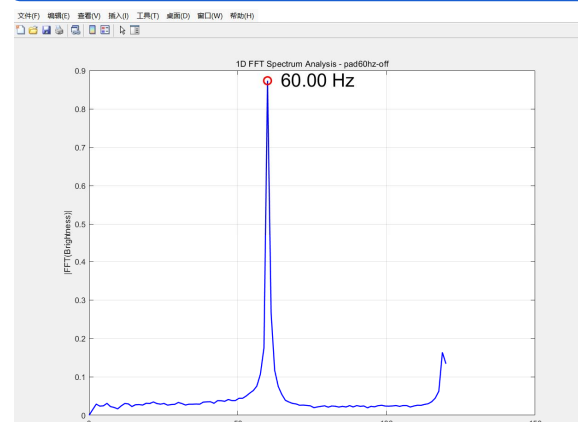
Results Analysis



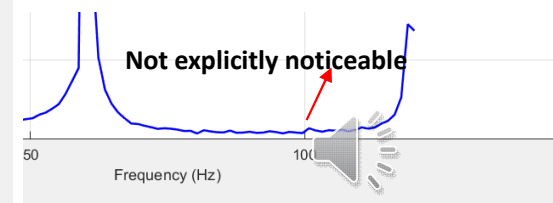
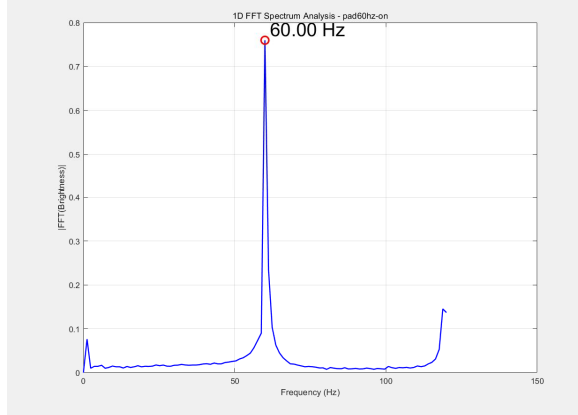
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pad_60hz_off



pad_60hz_on



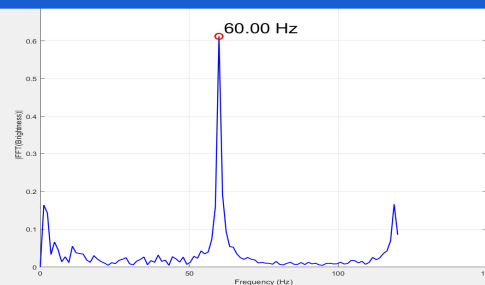
Results Analysis



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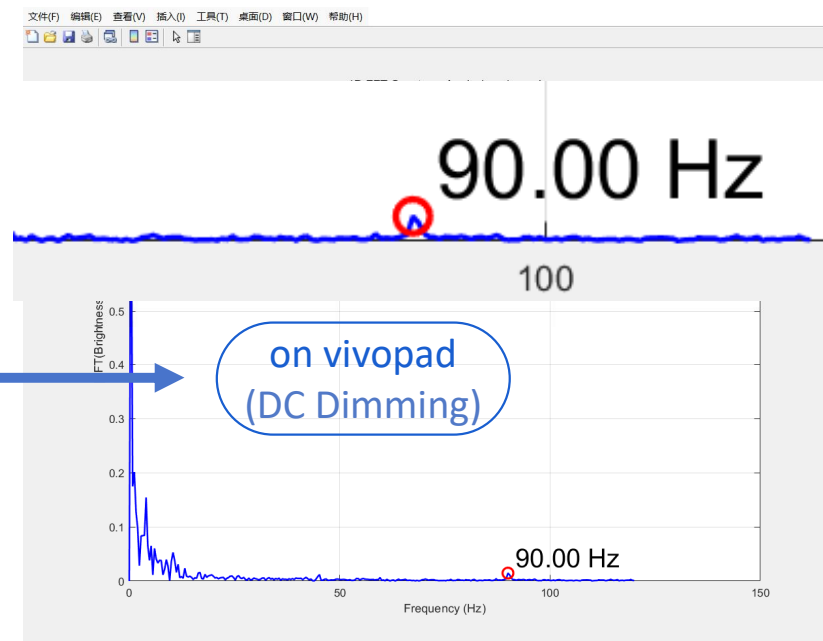
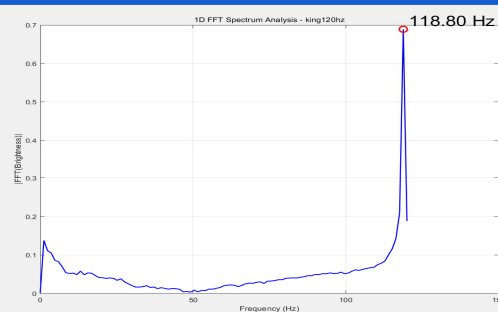
honor of king in 60hz



honor of king in 90hz

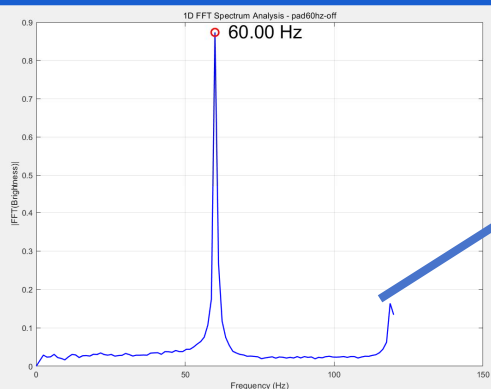


honor of king in 120hz

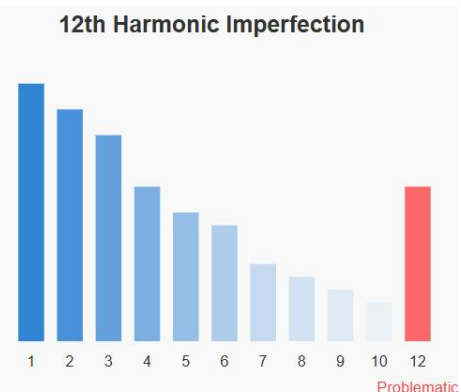


Interesting Phenomenon

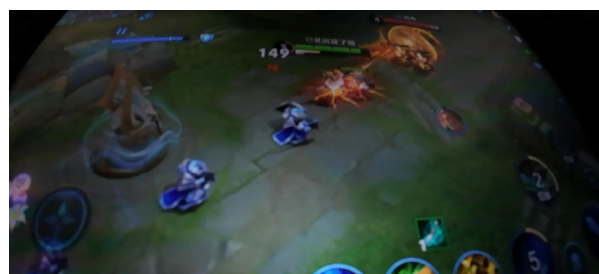
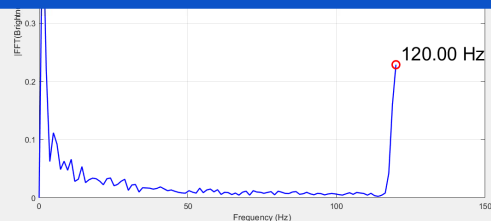
pad_60hz_on



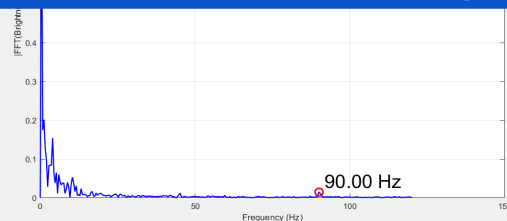
$\delta \otimes' \wedge \xi \in U \times \rightarrow \times$ 1: $\Psi \partial' U \supset U' \xi$ $\delta \times \sim \partial \wedge \nabla \rightarrow ' /$
 $\odot . I . \rightarrow \times \times U \cap$ $\sigma' U' \delta \sim + \partial$
 1440Hz PWM Signal



MatePad[PWM Dimming]

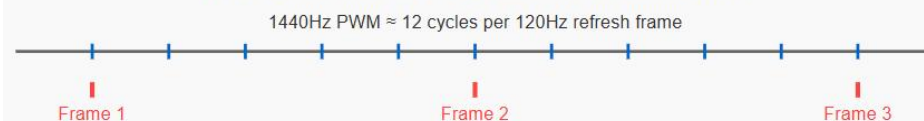


VivoPad[DC Dimming]



$\delta \otimes' \wedge \xi \in U \times \rightarrow \times$ 2: $\Upsilon \times \nabla U \downarrow ' \times$ $\delta \times \times U$
 $\sigma' \times \nabla \nabla \rightarrow \nabla \cap / \oplus \sim \rightarrow \ddagger \rightarrow \nabla \cap$ $4 \sim U' \ddagger U \nabla' \otimes$

Relationship with 120Hz Display Scanning



PWM Dimming (MatePad Pro)



DC Dimming (VivoPad)



Conclusion



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✓ Accurate Baseline & Screen Frequencies

- Validated FFT by detecting 100 Hz AC light flicker.
- Identified screen base frequencies (e.g., 60 Hz) & harmonics.

💡 PWM Dominance & Refresh Rate Decoupling

- Key Finding: For PWM-dimmed tablet (1440Hz PWM), measured flicker (~120Hz) often decoupled from set refresh rate (60/90Hz).
- ~120Hz peak likely aliased harmonic of PWM or panel's base drive.

⚠️ Nyquist Limit & Environmental Impact

- Demonstrated Nyquist limit: 144 Hz screen unmeasurable with 240fps camera.
- Showed ambient light interference & impact of screen brightness.

🎮 Real-World Implications & Dimming Tech

- Gaming tests confirmed PWM artifact/panel drive often dictates flicker on MatePad, irrespective of game's refresh setting.
- Comparison with DC-dimmed device highlighted different flicker profiles.

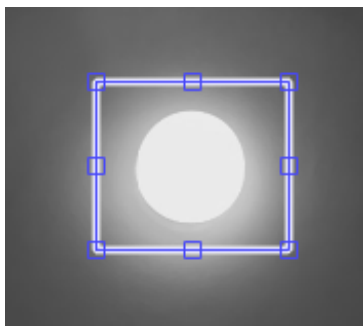
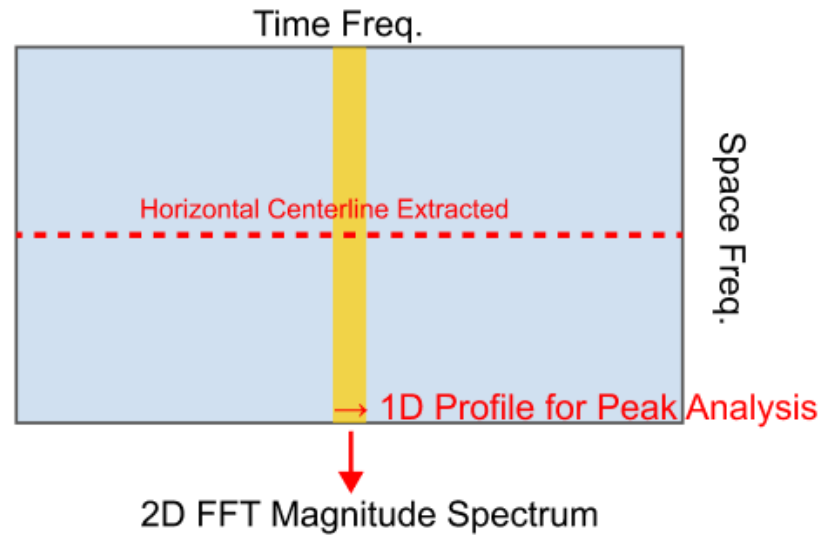


Explanation - 2D FFT

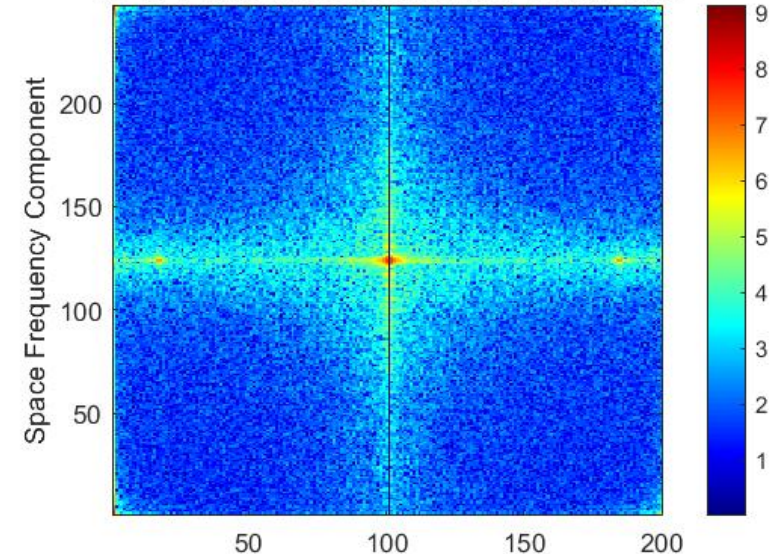


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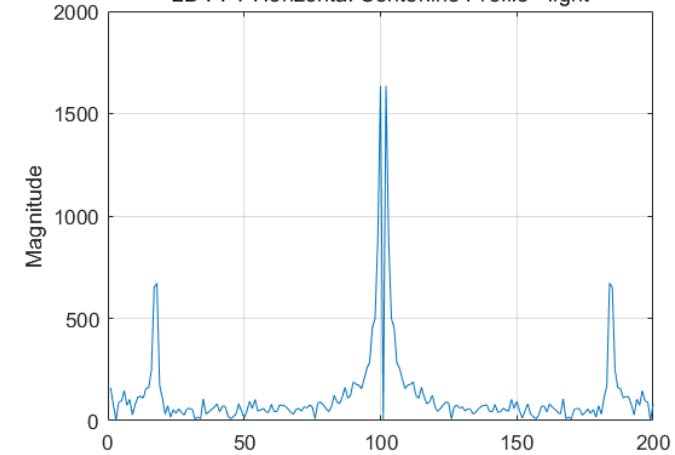
2D FFT Centerline Extraction



2D FFT Magnitude Spectrum (Log Scale) - light



2D FFT Horizontal Centerline Profile - light



Reference



1. $\oplus$. $\odot \times \S \times \nabla \cup \rightarrow$, ∇ . $* \dagger \nabla \S \cup \times \nabla \dagger$, $\S$. $\Omega \cup \partial \cup \times$, $\odot$. $\S \rightarrow \wedge ' \times \nabla \dagger$, $*$.
 $\Psi \wedge \nabla \cup \times ' \dagger \times \nabla \subset *$. $\odot \subset \wedge ' \in \rightarrow \S \cup \rightarrow$, " $\S \rightarrow \cap \in \S \supset \downarrow \rightarrow ' \leftarrow \cup \sim \subset \cup \S \cup ' \S \rightarrow \wedge \nabla$
 $\dagger \times \rightarrow \nabla \cap \in \rightarrow \cap \in - \times ' \cup \cup \subset \rightarrow \partial \times \cap \rightarrow \nabla \cap$," 2017 $\Psi \nabla \S \cup \sim \nabla \times \S \rightarrow \wedge \nabla \times \downarrow$
 $\S \wedge \nabla \supset \cup \sim \cup \nabla ' \cup \wedge \nabla * \wedge \subset \cup \sim \nabla \Upsilon \wedge \exists \cup \sim \S \otimes \times \S \cup \partial \times (* \Upsilon \S)$, $\S \downarrow \dagger \uparrow -$
 $\Omega \times ' \wedge ' \times$, $\Upsilon \wedge \partial \times \nabla \rightarrow \times$, 2017, ". 1-4, $\subset \wedge \rightarrow$:
10.1109/* $\Upsilon \S$.2017.7974448.
2. $\mathfrak{M} \rightarrow \leftarrow \rightarrow ' \cup \subset \rightarrow \times$. " $\Omega \otimes ' \dagger \rightarrow \times \S - \S \in \times \nabla \nabla \wedge \nabla \times \times \partial ' \downarrow \rightarrow \nabla \cap \S \in \cup \wedge \sim \cup \partial$.".
 $\odot \ddagger \times \rightarrow \downarrow \times ' \downarrow \cup$: $\in \S \S ' \times : // \cup \nabla . \exists \rightarrow \leftarrow \rightarrow ' \cup \subset \rightarrow \times . \wedge \sim \cap / \exists \rightarrow \leftarrow \rightarrow / \Omega \otimes ' \dagger \rightarrow \times \S - \S \in \times$
 $\nabla \nabla \wedge \nabla _ \times \times \partial ' \downarrow \rightarrow \nabla \cap _ \S \in \cup \wedge \sim \cup \partial$.
3. $\delta \dagger \times \exists \cup \rightarrow \S \wedge \nabla \times \dagger \partial \cup \sim \S \dagger \times \rightarrow \nabla \cup \times \times \mathfrak{h} \sim \wedge \dagger '$. " $\delta \mathfrak{M} \odot \mathfrak{M} \sigma \Psi * \times \S \cup \Upsilon \times \subset \Upsilon \sim \wedge$
13.2 $\rightarrow \nabla ' \in \S ' \cup ' \rightarrow \supset \rightarrow ' \times \S \rightarrow \wedge \nabla \times$ ".
 $\odot \ddagger \times \rightarrow \downarrow \times ' \downarrow \cup$: $\in \S \S ' \times : // ' \wedge \nabla \times \dagger \partial \cup \sim . \in \dagger \times \exists \cup \rightarrow . ' \wedge \partial / ' \nabla / \S \times ' \downarrow \cup \S \times / \partial \times \S \cup ' \times \subset$
 $- ' \sim \wedge - 13 - 2 / \times ' \cup \times /$.
4. ∇ ($\Omega \exists \rightarrow \S \S \cup \sim$), $@ \sigma \mathfrak{L} ' \downarrow \wedge \sim \cup \delta \mathfrak{P} \Omega \mathfrak{P} \Upsilon$," 3840 $\delta \odot \Upsilon \mathfrak{M} *$
 $\oplus \rightarrow \partial \partial \rightarrow \nabla \cap, \in \wedge \nabla \wedge \sim 90$ ".
 $\odot \ddagger \times \rightarrow \downarrow \times ' \downarrow \cup$: $\in \S \S ' \times : // \cup \nabla . \exists \rightarrow \leftarrow \rightarrow ' \cup \subset \rightarrow \times . \wedge \sim \cap / \exists \rightarrow \leftarrow \rightarrow / \Omega \otimes ' \dagger \rightarrow \times \S - \S \in \times$
 $\nabla \nabla \wedge \nabla _ \times \times \partial ' \downarrow \rightarrow \nabla \cap _ \S \in \cup \wedge \sim \cup \partial$.

Thanks for your Watching



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